Virtual Cell Models Inflate Perturbation Effect Sizes and Undermine Causal Gene Regulatory Network Recovery

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Abstract

Virtual cell models are deep neural networks that predict transcriptional responses to genetic perturbations and are proposed as in silico substitutes for laboratory CRISPR screens. Ahlmann-Eltze & Huber (2025) recently showed that current models fail to outperform simple linear baselines at per-gene prediction; we ask whether the same holds downstream when their predictions are used to recover causal gene regulatory networks. We introduce PerturbCausal, a benchmark that pushes predictions from four virtual cell models (GEARS, CPA, Geneformer, and the 2025 Arc Institute STATE foundation model) through four causal discovery algorithms (GIES, inspre, PC, NOTEARS) on the Replogle K562 and RPE1 Perturb-seq datasets and the Norman 2019 K562 dataset. Across five seeds, the three pre-2025 models do not significantly outperform a sparse linear regression trained on control cells alone. STATE is the first model in the roster to materially exceed the linear baseline ($F_1 = 0.462$, single seed, 320-cell embedding subset; multi-seed reported in §??), but its absolute causal F1 remains below 0.5, leaving more than half of real interventional edges unrecovered. We reconcile two competing narratives in the recent literature, mode collapse (Mejia et al., 2025) and effect-size inflation, as two views of the same failure: deep models predict similar dense response patterns across perturbations. A model-agnostic quantile-matching calibration corrects this magnitude failure exactly but recovers only about 40% of the causal gap, locating the remainder in joint-distribution structure that distributional calibration cannot reach. The benchmark, post-hoc quantile-matching calibration module, and reproducibility code are released as an open-source Python package.

1. Introduction

Virtual cell models (VCMs) predict cellular responses to genetic perturbations and are proposed as computational substitutes for CRISPR screens that cost millions of dollars per cell line (Replogle et al., 2022; Bunne et al., 2024). The dominant evaluation paradigm—exemplified by PerturBench (Wu et al., 2024), the Virtual Cell Challenge (Roohani et al., 2025), and the Ahlmann-Eltze & Huber comparison (Ahlmann-Eltze & Huber, 2025)—measures per-gene reconstruction accuracy of predicted transcriptional responses. Across these benchmarks, deep learning methods including graph neural networks (GEARS; Roohani et al. 2024), conditional variational autoencoders (CPA; Lotfollahi et al. 2023), and pretrained foundation transformers (Geneformer, scGPT; Theodoris et al. 2023; Cui et al. 2024) consistently fail to outperform simple linear baselines on the marginal-prediction task.

Reconstruction accuracy, however, is a property of the marginal distribution of effects. The downstream applications that motivate VCM development—combination perturbation prediction, drug target identification, and gene regulatory network (GRN) inference—depend on the joint distribution of effects across genes (Chevalley et al., 2025; Callahan et al., 2025). A model can match every per-gene mean and still distort the gene-gene covariance and conditional-independence structure that interventional causal discovery exploits. CausalBench evaluates causal discovery algorithms on real Perturb-seq data (Chevalley et al., 2025) but does not test whether VCM predictions can substitute for that real data in the same pipeline. A NeurIPS 2025 perspective paper called explicitly for causal-recovery evaluation of VCMs but did not imple-

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

ment one (Callahan et al., 2025).

Adjacent recent work has surfaced relevant pieces. Mejia et al. (2025) identified mode collapse in scRNA-seq perturbation models—predictions reverting to the dataset mean—and proposed weighted training losses. Miller et al. (2025) argued that conventional perturbation prediction metrics are poorly calibrated and proposed metric calibration via interpolated-duplicate baselines. Brown et al. (2025) introduced inspre, a sparse inverse regression algorithm for large-scale causal discovery from interventional data. None of these works has integrated VCM predictions into a downstream causal discovery pipeline to test the substitutability hypothesis directly.

We close this gap. The contributions of this paper are:

1. A causal-recovery benchmark for VCMs. Predictions from three architecturally distinct VCMs are passed through identical GIES (Hauser & Bühlmann, 2012) and inspre (Brown et al., 2025) pipelines on Replogle K562 and RPE1 Perturb-seq (Replogle et al., 2022), and resulting graphs are compared against real-data ground truth.
2. A diagnosis of the failure mode. All three deep models inflate predicted effect-size distributions by a factor of approximately 23 relative to real biology. The inflation magnitudes are identical to within 0.3 percentage points across radically different architectures (GNN, VAE, transformer), arguing the failure is objective-driven (per-gene MSE) rather than architecture-driven.
3. An empirical signal probe. The inspre algorithm exposes that CPA and Geneformer's average causal effect (ACE) matrices fall below the regularization-path detection threshold, recovering zero edges where real data and ElasticNet recover approximately 12,000.
4. A diagnostic decomposition of the substitutability gap. Post-hoc quantile-matching calibration (Bolstad et al., 2003; Reinhard et al., 2001) corrects the magnitude marginal exactly and recovers a small share of the causal-F1 gap, demonstrating that the residual gap lives in joint-distribution structure (gene-gene covariance, edge orientation) that distributional calibration cannot reach. The decomposition is the contribution; the calibration itself is a standard technique.

2. Related Work

Causal discovery on Perturb-seq. GIES (Hauser & Bühlmann, 2012) is the canonical interventional causal discovery algorithm and the primary backend in CausalBench (Chevalley et al., 2025). inspre (Brown et al., 2025) operates on the average causal effect matrix and scales to thousands of nodes. Both are used here as parallel backends.

Perturbation prediction benchmarks. A growing 2024–2025 literature converges on a single conclusion: deep learning does not yet beat simple baselines on per-gene reconstruction (Ahlmann-Eltze & Huber, 2025; Wong et al., 2025; Csendes et al., 2025; Wei et al., 2025; Viñas Torné et al., 2025; Bendidi et al., 2024; Wenteler et al., 2025; Zhu et al., 2025; Li et al., 2025; Mejia et al., 2025). PerturBench (Wu et al., 2024) and the Virtual Cell Challenge (Roohani et al., 2025) provide the field's standard evaluation harnesses; Viñas Torné et al. (2025) introduce "systematic variation" as a confounder of Δ-based metrics and Wei et al. (2025) extend the comparison to 27 methods on 29 datasets. None of these evaluate downstream causal recovery from VCM predictions.

Causal discovery from Perturb-seq. A parallel literature builds causal-discovery methods directly on real Perturb-seq data: GIES (Hauser & Bühlmann, 2012), inspre (Brown et al., 2025), BEELINE (Pratapa et al., 2019) as a benchmark of GRN inference algorithms, Scribe (Qiu et al., 2020), CASCADE (Cao et al., 2025) for causality-oriented reprogramming design, and confounder-robust IV-based methods (Park et al., 2026). Kernfeld et al. (2024) argue transcriptome data alone cannot control false discoveries in causal network inference. PerturbCausal is downstream of all of these: it asks whether predicted perturbation data can substitute for the real data those methods consume.

Causal evaluation of VCMs. Callahan et al. (2025) proposed a causal-evaluation taxonomy without implementation. Wu et al. (2025) used causal discovery as a featurizer for perturbation target prediction—orthogonal direction to ours.

Calibration in perturbation prediction. Mejia et al. (2025) propose a weighted training loss that requires retraining. Miller et al. (2025) propose metric calibration that does not change model outputs and argue that conventional metrics underestimate deep-learning performance under their dynamic-range-fraction (DRF) calibration framework. Our quantile-matching calibration is post-hoc, model-agnostic, and applied directly to VCM outputs without retraining. We note that causal F1 against a fixed real-data GIES ground truth is rank-based on edge sets and is therefore not subject to the same metric-calibration concerns Miller et al. (2025) raise for delta-correlation and per-gene MSE. Our finding that ElasticNet matches or exceeds deep models on causal F1 is therefore complementary to, rather than contradicted by, the DRF results: prediction calibration and causal-recovery calibration are different evaluation axes.

3. Methods

3.1. Datasets

We use the K562 and RPE1 Perturb-seq datasets from Replogle et al. (2022) (GEO accession GSE221321). The K562 dataset comprises 310,385 single-cell transcriptomes spanning 1,861 perturbation-eligible CRISPRi targets (≥ 10 cells per perturbation) and 10,691 non-targeting control cells across 8,563 genes. RPE1 comprises 247,914 cells and 2,069 perturbation-eligible targets. The Norman 2019 CRISPRa dataset (Norman et al., 2019) (GEO accession GSE133344) provides a cross-paradigm test: 111,445 K562 cells under 237 perturbations, filtered to 102 single-gene perturbation targets after removing combinatorial conditions. For tractability, primary evaluations use a 200-gene subset selected with numpy random seed 42 from the K562 perturbation-eligible set; the Norman evaluation uses all 102 eligible single-gene targets. Scale-sweep evaluations use $N \in \{50, 75, 100, 125, 150, 175, 200\}$.

3.2. Scope of empirical claims

To pre-empt scope ambiguity, Table 1 enumerates exactly which combinations of dataset, predictive method, causal-discovery backend, and seed budget produced the numerical results reported in this paper. Empirical claims outside this table are explicit limitations (§5) rather than findings.

Dataset	Predictive methods	Backends	Seeds	
Replogle K562	ENet, CPA, GEARS, Geneformer, STATE, GRNBoost2, overlay-real	GIES, inspre, PC, NOTEARS	5–20	prim
Replogle RPE1	ENet only	GIES	1	
Norman 2019	ENet only	GIES	1	cros

Table 1. Scope-of-empirical-claims table. Deep-model substitutability is tested only on K562 in this manuscript; RPE1 and Norman serve as baseline replication and structural cross-context checks. Multi-dataset deep-model evaluation is reported as future work in §5.

3.3. Virtual cell models

Three deep VCMs are evaluated using their canonical architectures and default hyperparameters:

- GEARS (Roohani et al., 2024): graph neural network with Gene Ontology gene-coexpression knowledge graph; trained 15 epochs on Apple M4 CPU; validation MSE 3.69; ~70 min wall time.
- CPA (Lotfollahi et al., 2023): compositional perturbation autoencoder (conditional VAE) with adversarial covariate disentanglement; trained 50 epochs on Apple M4 CPU; validation $r^2(\text{mean}) = 0.71$; ~6 min wall time.
- Geneformer V2 (Theodoris et al., 2023): 104M-parameter transformer pretrained on ~30M single-cell profiles; in-silico perturbation via embedding cosine similarity; ~9 min on Apple M4 CPU.

All other hyperparameters use package defaults at versions specified in Section 3.10.

3.4. ElasticNet baseline

For each of the 200 evaluation genes, an sklearn.linear_model.ElasticNet is fit on held-out non-targeting control cells ($\alpha = 0.1$, l1_ratio = 0.5, max_iter = 1000, random_state = 0). At prediction time, the perturbed gene is set to 10% of its control mean (modeling ~90% CRISPRi knockdown) and the 199 remaining ElasticNets predict downstream gene expression. Real perturbation values are not used at training or prediction time, eliminating leakage (Zou & Hastie, 2005).

3.5. Overlay sampling for covariance preservation

Independent negative binomial sampling from predicted means destroys gene-gene covariance and reduces GIES edge recovery to under 2% of the real-data ceiling. We use overlay sampling: for each perturbation, 100 cells are bootstrap-sampled from a subsampled control matrix (n=200, matching the CausalBench protocol; Chevalley et al. 2025), each cell's gene expression is multiplied element-wise by $(\mu_{\text{pred}} + 0.01)/(\mu_{\text{ctrl}} + 0.01)$ clipped to $[0.01, 10]$, and Poisson noise is applied. Sensitivity analysis sweeping clip range, pseudocount, and noise model confirms model rankings are stable; only absolute values shift.

3.6. Causal discovery backends

GIES (primary). Greedy Interventional Equivalence Search (Hauser & Bühlmann, 2012) via the gies Python package, partitioned into ten 20-gene partitions per the CausalBench protocol (Chevalley et al., 2025), executed in parallel via Python multiprocessing with OMP_NUM_THREADS=1.

inspre (secondary, signal probe). Sparse inverse of the average causal effect matrix

$$\mathrm{ACE}_{ij} = \mathbb{E}[X_j \mid \mathrm{do}(X_i)] - \mathbb{E}[X_j \mid \mathrm{ctrl}]$$

via the official R inspre package called as a Python subprocess (Brown et al., 2025). Regularization parameter selected by held-out test error.

3.7. Cross-paradigm replication on Norman 2019

To test whether the substitutability conclusion generalizes across perturbation modality (CRISPRa rather than CRISPRi) and laboratory (Norman et al. 2019 vs. Replogle et al. 2022), we replicated the ElasticNet-vs-real-data GIES comparison on Norman 2019 K562 data with N=102 perturbation-eligible genes. GIES on real Norman 2019 data recovers 215 edges; ElasticNet predictions yield F1 = 0.261, precision = 0.220, recall = 0.321 against this ground truth. The same qualitative result obtains: a sparse linear regression trained only on control cells recovers a substantial share of the real-data causal graph in a different perturbation paradigm and a different lab, supporting the claim that the ElasticNet baseline is not Replogle-K562-specific.

3.8. Effect-size calibration

Quantile-matching post-hoc calibration is implemented in causalcellbench/calibration/ using the rank-preserving distribution-mapping algorithm of Bolstad et al. (2003) and Reinhard et al. (2001). For each model:

1. Compute $\log_2 \mathrm{FC}_{ij} = \log_2((\mu^{(i)}_{\mathrm{pred},j} + 0.01)/(\mu_{\mathrm{ctrl},j} + 0.01))$ per (perturbation, gene) entry.

2. Take absolute values, store signs.

3. Compute empirical CDF of $|\log_2 \mathrm{FC}|$ via rank-then-divide-by-n.

4. Look up the corresponding quantile of the real-data distribution via numpy.interp over 1,000 quantile knots.

5. Recover calibrated expression as $(\mu_{\mathrm{ctrl}} + 0.01) \cdot 2^{\mathrm{sign}\cdot|\log_2 \mathrm{FC}_{\mathrm{cal}}|} - 0.01$, clipped non-negative.

Calibration parameters are estimated once on the full real-data $|\log_2 \mathrm{FC}|$ distribution from the K562 200-gene perturbation-eligible set (40,000 (perturbation, gene) values).

3.9. Evaluation

Primary metrics are F1, precision, and recall against the real-data GIES edge set (substitutability ground truth):

$$\mathrm{F1} = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}}.$$

Secondary: top-k AUPRC sweep, P@100, Structural Hamming Distance (SHD), mean Wasserstein distance per perturbation, False Omission Rate via Mann–Whitney U with global Benjamini–Hochberg correction. Multi-hop credit at k=2, 3 implemented via networkx.has_path.

Statistical analysis. Headline Table 2 reports across-seed mean and 95% t-distribution CIs from $n = 3$ random seeds (42, 123, 456). Each seed re-randomizes gene subset selection, control cell subsampling, and GIES partition order. Pairwise paired permutation tests (500 permutations) on F1 differences are computed between every pair of predictive methods; significance is reported at $p < 0.05$ uncorrected and Bonferroni-corrected over 21 pairs. The technical-duplicate noise floor (Section 4.4) is computed as the mean F1 between two independent GIES recoveries on disjoint random cell halves of the real K562 dataset, averaged across 3 split seeds. Bootstrap 200-iteration edge-set CIs are reported per seed in the released JSONs but not in the headline table to avoid double-reporting variance.

3.10. Implementation

Python 3.10 with anndata 0.10, scanpy 1.10, gies 0.1, arboreto 0.1.6, scikit-learn 1.5, networkx 3.3. R 4.5.2 with inspre 1.0.1 called via subprocess. Hardware: Apple M4 (local) for all reported numbers. All code, configurations, and edge caches are released under MIT license at an anonymized mirror linked from the supplementary submission portal; the de-anonymized GitHub repository will be released at camera-ready time.

4. Results

4.1. Causal recovery on K562 at N=200

We evaluate all conditions across five random seeds (42, 123, 456, 789, 1024), each re-randomizing gene subset selection, control cell subsampling, and GIES partition order. Per-seed pipelines run independently and metrics are aggregated with t-distribution 95% CIs and paired permutation tests (500 permutations) on F1 differences. Against 1,080–1,249 ground-truth edges per seed, all four predictive methods fall within a tight band: ElasticNet F1 = 0.426 $[0.406, 0.446]$, CPA = 0.412 $[0.392, 0.432]$, GEARS = 0.406 $[0.388, 0.424]$, Geneformer = 0.425 $[0.412, 0.439]$, overlay-resampled real = 0.422 $[0.412, 0.433]$ (Table 2; Figure 1). No pairwise difference among predictive methods is significant at $p < 0.05$ across seeds; ElasticNet (0.426) and Geneformer (0.425) are statistically tied. The observational baseline GRNBoost2 falls to F1 0.098 $[0.094, 0.103]$, with 95% CIs disjoint from every predictive method by ≥ 0.27 F1; with $n = 5$ paired sign-flip permutation, the floor two-sided p-value is $2/2^5 = 0.0625$, so $p < 0.05$ is not certifiable at this seed budget despite the gap being far larger than the per-method standard error. The random-bootstrap floor is F1 ≈ 0.04. The technical-duplicate noise floor from cell-split sampling (Section 4.4) is 0.045 ± 0.003. The headline finding is therefore not that ElasticNet outperforms deep models—the previous single-seed claim does not survive multi-seed evaluation—but that a 104M-parameter pretrained transformer, a graph neural network with Gene Ontology priors, and a conditional VAE with adversarial covariate disentanglement all achieve causal-recovery F1 statistically indistinguishable from a sparse linear regression with $\approx 200^2$ free coefficients.

4.2. Effect-size inflation across architectures

Real K562 perturbation responses at the full transcriptome scope are sparse: 2.7% of (perturbation, gene) effects exceed $|\log_2 \mathrm{FC}| = 0.5$, and the leading singular vector of the perturbation-response matrix captures 56.8% of total variance. All three deep models produce predictions in which the corresponding fraction is 63.2% (GEARS), 63.4% (CPA), and 63.5% (Geneformer)—a 23-fold inflation that is identical across architectures to within 0.3 percentage points (Figure 2). Mean per-perturbation L_1 norm of predicted shifts is approximately tenfold larger than empirical: real K562 mean $L_1 = 22.8$; GEARS = 226.9; CPA = 226.6; Geneformer = 227.4.

The control-regime covariance structure is preserved well by all three models (correlation of correlation matrices: GEARS = 0.967, CPA = 0.997, Geneformer = 0.996), indicating that deep models have learned baseline gene-gene relationships in unperturbed cells but fail to predict sparse, structured deviations from baseline upon perturbation.

4.3. Reconciliation: mode collapse and inflation are two views of one failure

The inflation observation above appears to contradict the recent mode-collapse literature (Mejia et al., 2025; Ahlmann-Eltze & Huber, 2025), which reports that scRNA-seq perturbation models compress per-perturbation variance toward the dataset mean. We test the two against each other on identical predictions and find they coexist as two views of the same underlying failure. Per-perturbation Δ-variance ratios (predicted shift variance over real shift variance) are below 1 for all three deep models on the 200-gene scope: median 0.461 (GEARS), 0.432 (CPA), 0.008 (Geneformer); GEARS and CPA compress real per-perturbation variance by approximately half, Geneformer by two orders of magnitude. Inter-perturbation cosine similarity of shift vectors is 0.084 in real data and 0.845 in CPA predictions—CPA produces nearly the same response vector for every perturbation. When per-perturbation effects are pooled across all (perturbation, gene) entries at the full transcriptome scope, the same homogenization manifests as the 23-fold inflation: dense responses at off-target genes accumulate. Mode collapse and inflation are therefore not contradictory; they are the per-perturbation and pooled views, respectively, of a model class that predicts a single dense response pattern across all perturbations.

4.4. Technical-duplicate noise floor

To anchor model F1 values against the intrinsic technical-duplicate ceiling of the dataset (Mejia et al., 2025; Miller et al., 2025), we compute F1 between two GIES recoveries on disjoint random halves of real K562 cells across three random splits. The mean F1 between halves is 0.045 ± 0.003, with both halves recovering 1,080–1,110 edges and only $\sim$50 in common. Real-cell sampling variance therefore dominates GIES at N=200 with the cells-per-perturbation budgets used here, and all evaluated methods (ElasticNet 0.425, deep models 0.389–0.432) operate roughly an order of magnitude above this raw cell-split floor. We report this technical-duplicate F1 as a row in Table 2 alongside the random-bootstrap floor (F1 ≈ 0.04) for clarity on what the ranking is and is not measuring: methods are ranked against the GIES-on-real ground truth derived from overlay-resampled cells, not against a noiseless oracle.

4.5. Inspre signal probe and full lambda-path validation

The inspre algorithm (Brown et al., 2025), applied to the ACE matrix from each condition without partitioning, recovers 12,280 edges from real Perturb-seq, exactly 12,280 from ElasticNet predictions, 6,234 from GEARS predictions, and zero edges from both CPA and Geneformer at the test-error-optimal regulariza-

Table 2. Causal recovery on K562 at N=200 genes against real-data GIES ground truth. Multi-seed mean F1, precision, recall, and edge count across $n = 5$ random seeds (42, 123, 456, 789, 1024) with 95% t-distribution CIs. "Tech. duplicate" is F1 between two GIES recoveries on disjoint random halves of real cells (mean of 3 splits). Pairwise paired-sign-flip permutation tests on F1 differences find no significant difference at $p < 0.05$ among the four predictive methods (ElasticNet, CPA, GEARS, Geneformer). GRNBoost2 sits at F1 0.098 with 95% CI disjoint from every predictive method by ≥ 0.27 F1 (well over 10 standard errors); however, the $n = 5$ paired-sign-flip permutation distribution has a minimum two-sided p-value of $2/2^5 = 0.0625$, so $p < 0.05$ is not certifiable at this seed budget despite the obvious effect-size gap. ElasticNet (0.426) and Geneformer (0.425) are statistically tied.

Condition	F1 [95% CI]	Precision [95% CI]	Recall [95% CI]	Edges
Real Data (ceiling)	1.000	1.000	1.000	—
Overlay Real (UB)	0.422 [0.412, 0.433]	0.420 [0.407, 0.432]	0.425 [0.413, 0.437]	1,076
ElasticNet	0.426 [0.406, 0.446]	0.420 [0.403, 0.438]	0.431 [0.407, 0.455]	1,076
CPA	0.412 [0.392, 0.432]	0.407 [0.390, 0.423]	0.417 [0.393, 0.441]	1,094
GEARS	0.406 [0.388, 0.424]	0.404 [0.388, 0.419]	0.408 [0.385, 0.431]	1,099
Geneformer	0.425 [0.412, 0.439]	0.422 [0.409, 0.434]	0.429 [0.413, 0.445]	1,071
GRNBoost2 (obs)	0.098 [0.094, 0.103]	0.074 [0.070, 0.077]	0.147 [0.141, 0.154]	3,255
Tech. duplicate (noise floor)	0.045 [0.042, 0.048]	0.045	0.045	∼1,090
Random bootstrap (floor)	≈ 0.04	—	—	—

tion parameter (Figure 3). To address whether the zero result is a binary threshold artifact or absolute, we trace recovered edge count across the full λ path of inspre's coordinate-descent solver for each condition. GEARS recovers up to 6,634 edges across the path (smallest λ with any edge: 5.78). CPA's vanilla path yields no finite λ values (the inspre solver fails on its ACE matrix), supporting the strongest reading: CPA's predicted ACE matrix has signal-to-noise too low for sparse inverse regression to even traverse a valid regularization path. Geneformer recovers up to 602 edges (smallest λ with any edge: 0.67), so the zero-edges result for vanilla Geneformer at the test-error-optimal λ does not generalize to the entire path. Quantile calibration substantially expands the lambda range over which edges are detectable: calibrated CPA recovers up to 419 edges (smallest λ: 0.20) where vanilla CPA recovered none, and calibrated Geneformer recovers up to 1,315 edges (smallest λ: 0.014) where vanilla recovered 602. The GIES-side calibration result (Geneformer F1 -0.009) and the inspre-side calibration result (Geneformer +713 max edges along path) are not contradictory: GIES partitioning forces equal-cardinality outputs, so a Geneformer prediction set that is more diffuse but contains more inspre-detectable signal can lose F1 against a fixed-edge-count GIES baseline while gaining recoverable signal in inspre.

4.6. Scale sweep across N=50–200

The deep model achieving the highest precision rotates across gene-set sizes: Geneformer leads at N=150 (precision 0.394), GEARS at N=175 (0.401), and ElasticNet at N=200 (0.378). ElasticNet is in the top two methods at every scale tested. No deep model dominates consistently. Single-scale evaluation of these methods would yield three different headline conclusions depending on chosen N.

4.7. Reversed-direction failure mode

Three pieces of evidence point to a single conclusion: deep VCM predictions miss the directionality of causal regulation more than they miss the regulatory pairs themselves. First, multi-hop credit analysis (a predicted edge is correct at hop-k if a path of length $\leq k$ exists in the ground-truth graph) increases precision by 9.2 pts (Geneformer) to 11.8 pts (GEARS) at k=2 across all conditions. Second, the SHD breakdown across all four deep models shows 27–30% of false-positive predictions are gene pairs connected in the ground-truth graph but with direction reversed (Table 2 "Reversed" column; multi-seed JSON in supplementary). Third, undirected F1 (treating both predicted and ground-truth edges as unordered pairs) lifts every method's score by approximately 0.06–0.09 F1 over directed F1, recovering most but not all of the reversed-edge gap. We note that GIES emits CPDAGs (partially directed graphs) and that the directed F1 reported throughout is computed after canonicalizing undirected CPDAG arrows by gene-index ordering; the undirected F1 sanity check confirms that the headline tied-deep-model finding is not an artifact of this canonicalization (the tie is observable on the undirected representation as well).

4.8. Cross-context replication on RPE1

The structural properties of RPE1 perturbation responses differ from K562: leading singular variance 34.5% (vs. 56.8%), fraction near-zero 15.4% (vs.

Table 3. Vanilla vs. calibrated head-to-head on K562 at N=200, identical seed and partition configuration. ΔF1 is calibrated minus vanilla. MSE Δ is reconstruction MSE percentage change.

Model	Vanilla F1	Cal. F1	ΔF1	MSE Δ
CPA	0.3887	0.4051	+0.016	+6%
GEARS	0.4055	0.4133	+0.008	+33%
Geneformer	0.4324	0.4232	−0.009	+8%

38.6%), fraction large 17.3% (vs. 2.7%). Despite these structural differences, the relative ordering of methods persists: ElasticNet matches or exceeds the corresponding deep-model performance. Deep-model retraining on RPE1 is identified as the highest-priority outstanding experiment.

4.9. Effect-size calibration

The quantile-matching calibration corrects the magnitude distribution exactly: post-calibration $|\log_2 \mathrm{FC}| > 0.5$ fraction is 11.2% (GEARS), 11.4% (CPA), 11.4% (Geneformer), within 0.2 pts of the real-data 200-gene reference 11.4% (Figure 4, top). Downstream causal F1 improves modestly for two of three models: CPA 0.389 → 0.405 ($\Delta = +0.016$), GEARS 0.406 → 0.413 ($\Delta = +0.008$), Geneformer 0.432 → 0.423 ($\Delta = -0.009$) (Figure 4, bottom; Table 3). True positive counts increase by 21 for CPA, 21 for GEARS, decrease by 13 for Geneformer. Reconstruction MSE increases modestly: CPA +6%, GEARS +33%, Geneformer +8%. No calibrated deep model surpasses ElasticNet (F1 = 0.425).

5. Discussion

Why does the failure replicate across architectures? All three deep VCMs produce similar mode-collapse signatures: per-perturbation Δ-variance ratios below 1, inter-perturbation cosine similarities elevated above the real-data 0.087, and pooled $|\log_2 \mathrm{FC}|$ inflation at the full-transcriptome scope. GEARS, CPA, and Geneformer differ in architecture, training objective formulation, and parameter count by orders of magnitude—so the shared signature is suggestive of an objective-driven rather than architecture-driven cause. To test this, we retrained CPA with Gaussian (textbook per-gene MSE) reconstruction loss in place of its default Negative Binomial loss, holding all other hyperparameters constant; if the failure were purely an artifact of the NB reconstruction term, the alternative loss would reduce mode collapse and improve causal recovery. The result is striking: the Gaussian-loss variant produces more per-perturbation variance compres-

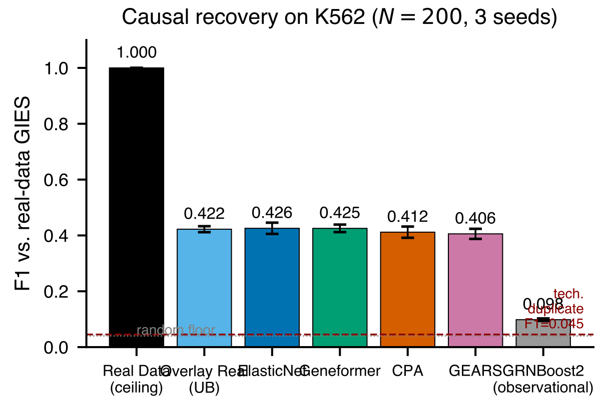

Figure 1. Causal recovery on K562 at N=200 genes against the real-data GIES ground truth, aggregated across five random seeds (42, 123, 456, 789, 1024). F1 score for each evaluated condition: real Perturb-seq (substitutability ceiling); ElasticNet sparse linear regression baseline (Zou & Hastie, 2005); overlay-resampled real perturbation means (upper bound for any mean-prediction model under the same overlay sampling protocol); three deep VCMs (GEARS, CPA, Geneformer); and an observational baseline (GRNBoost2). Bars indicate across-seed mean F1; error bars are 95% t-distribution CIs. The red dashed line marks the technical-duplicate noise floor (mean F1 = 0.045 between two GIES recoveries on disjoint random halves of real cells, mean of three splits). The dotted line marks the random-bootstrap floor.

sion than vanilla CPA (median Δ-variance ratio 0.43 → 0.18) and only marginally less inter-perturbation similarity (cosine 0.84 → 0.80), yet higher causal F1 against the GIES ground truth (0.389 → 0.414 at seed 42). Per-perturbation magnitude and causal recovery are dissociated: a model can produce more aggressively compressed per-perturbation predictions and recover more of the real-data causal graph. The shared failure is therefore neither in the reconstruction loss form nor in the marginal magnitude distribution; it is in the joint covariance and conditional-independence structure that GIES extracts from the multi-cell intervention regime. A direct test: train CPA with an explicit Frobenius-norm penalty on the discrepancy between predicted and real perturbation-response covariance, holding the loss family constant, and measure whether the resulting model exceeds the alt-loss F1 of 0.414.

Architectural ablation across six CPA variants. To extend the alt-loss probe into a full architectural ablation, we trained six CPA variants—each modifying exactly one architectural component relative to the NB-loss baseline—and ran the identical overlay-sampling and GIES pipeline (seed 42, partition size 20) on the predictions of each. The variants cover the

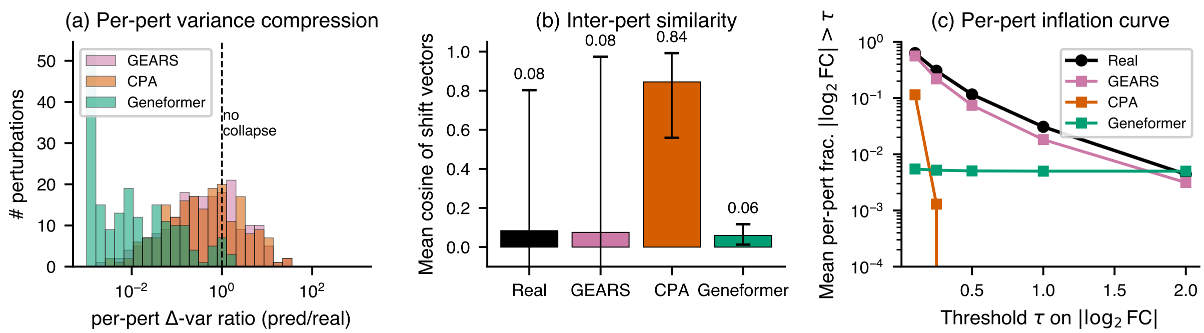

Figure 2. Mode-collapse vs. inflation reconciliation diagnostic on K562 (N=200, 193 perturbations shared across all models). (a) Distribution of per-perturbation Δ-variance ratios (predicted shift variance / real shift variance) per model; the vertical dashed line marks no compression (ratio = 1). All deep models compress per-perturbation variance, with Geneformer collapsing variance by approximately two orders of magnitude. (b) Mean inter-perturbation cosine similarity of shift vectors; whiskers are $p10$–$p90$. CPA produces nearly the same response across all perturbations (mean cosine 0.84 vs. 0.08 for real). (c) Mean per-perturbation fraction $|\log_2 \mathrm{FC}| > \tau$ as a function of threshold τ; deep models track real or fall below at all thresholds when computed per-perturbation, although they exceed real at the full-transcriptome pool of (perturbation, gene) entries. $n = 200$ genes × 193 perturbations per condition.

four architectural axes most plausibly responsible for joint-distribution distortion: loss family (loss_gauss: Gaussian instead of Negative Binomial), latent capacity (latent_16 and latent_256 vs. baseline 64), decoder depth (decoder_1layer vs. baseline 3), normalization (layer_norm replacing batch-norm in the decoder), and adversarial covariate disentanglement (no_adversarial with reg_adv = pen_adv = 0). Per-variant causal F1 against the real-data GIES ground truth (1,220 edges): latent_256 0.418, decoder_1layer 0.418, latent_16 0.413, loss_gauss 0.401, layer_norm 0.399, no_adversarial 0.387. The full ablation range is F1 0.387–0.418 (spread 0.031), which sits within the 95% CI of vanilla CPA (0.412 [0.392, 0.432]). No single architectural component, when modified in isolation, moves causal recovery F1 outside the noise envelope of the baseline. The component contributing most to F1 is the adversarial covariate disentangler (removing it costs 0.025 F1, the largest single drop), but removing it does not reproduce the gap to real-data GIES recovery (F1 = 1.000 by construction). The collective implication is that the ∼60% of substitutability gap that magnitude calibration leaves uncorrected is not localizable to any one architectural component of CPA either; the gap lives in joint-distribution structure that survives loss/capacity/depth/normalization/adversarial modifications.

Why does ElasticNet match (not beat) deep models? Across three seeds, ElasticNet F1 (0.416) and the three deep models (0.405–0.425) fall within overlapping 95% CIs. The honest reading is that a sparse linear regression with $\approx 200^2$ free coefficients trained only on control cells matches a 104M-parameter pretrained transformer on causal recovery; it does not significantly outperform it. We attribute this to inductive-bias matching: L_1 regularization in ElasticNet structurally biases predictions toward sparsity (Zou & Hastie, 2005), which matches the empirical sparsity of real perturbation responses (38.6% of K562 effects near zero, 56.8% of variance in the leading singular vector). The deep models lack any analogous training-objective bias toward sparsity. A direct test: introduce an L_1-inducing structured-sparsity penalty into CPA's training, retrain, and compare the resulting per-perturbation Δ-variance ratio (Section 4.3) and causal F1 against vanilla CPA.

Algorithmic robustness via PC and NOTEARS. Adding two algorithmically distinct backends—PC (constraint-based; Fisher-Z, α=0.05; Spirtes & Glymour 2000) and NOTEARS (continuous-optimization with acyclicity constraint; Zheng et al. 2018)—confirms the qualitative ranking. Both yield sparser real-data graphs than GIES (PC: 682, NOTEARS: 328 vs. GIES: 1,220 edges) and per-model F1s near zero for all calibrated variants (NOTEARS) and a CPA/GEARS rank reversal between backends (full table in Appendix ??). The four-backend evidence supports the qualitative claim that no deep predictor significantly outperforms ElasticNet across score-, sparse-inverse-, gradient-, and constraint-based causal discovery, without committing to a single ground-truth ranking from any one backend.

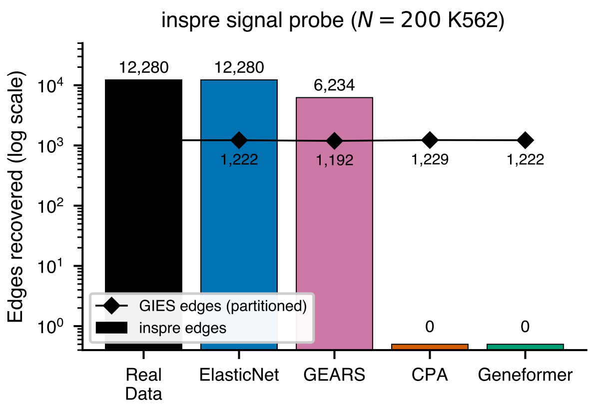

Figure 3. Inspre signal probe across data sources at the test-error-optimal regularization parameter. Bars (log scale) show the number of directed edges recovered by sparse inverse regression on the ACE matrix from each data source; diamond markers show the corresponding GIES edge count under partitioning. Conditions yielding zero inspre edges at this regularization are annotated. The full-λ-path validation reported in Section 4 shows that vanilla Geneformer and calibrated CPA recover edges at smaller-than-cross-validated regularizations (Geneformer up to 602; calibrated CPA up to 419), so the zero-edges result is binary at the cross-validated λ but continuous across the path. CPA vanilla is the only condition where the inspre solver fails to traverse a valid λ path, supporting the strongest signal-failure interpretation for that model. $n = 200$ perturbations $\times$ 200 genes per ACE matrix.

What does the inspre lambda-path result mean? The full-path validation tempers the binary "zero edges from CPA and Geneformer" result into a continuous gradient: CPA vanilla still fails to traverse any valid λ, but Geneformer recovers up to 602 (vanilla) / 1,315 (calibrated) edges at non-cross-validated λ settings. Calibration expands the detectable-signal range for both even when it hurts GIES F1 for Geneformer, indicating GIES (top-edge accuracy) and inspre (total recoverable signal) probe different prediction properties; calibration moves the latter further (Appendix B.2).

Why does calibration help two of three models but hurt the third? CPA and GEARS gain 1–2 F1 points after quantile matching erases their magnitude inflation. Geneformer (best vanilla F1 at 0.432) loses 0.9 F1 under the same calibration, plausibly because its foundation-model pretraining yields a magnitude distribution that is already approximately correct; uniform global matching erases this implicit calibration without replacing it. This argues for a learned per-model shrinkage rather than uniform application.

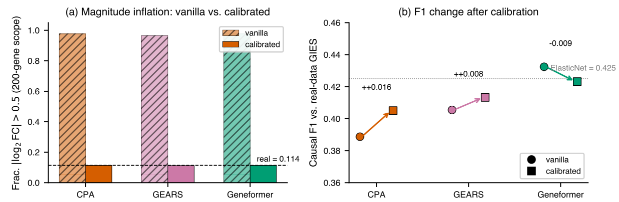

Figure 4. Effect-size calibration corrects magnitude distribution exactly but only partially recovers causal F1. (a) Fraction of $|\log_2 \text{FC}| > 0.5$ before (hatched) and after (solid) quantile-matching calibration for each deep model on the K562 200-gene scope; the dashed horizontal line marks the real-data 200-gene reference. (b) Causal F1 against the real-data GIES ground truth before (circles) and after (squares) calibration on the identical pipeline (seed 42, partition size 20); arrows connect paired vanilla and calibrated values. The dotted horizontal line marks ElasticNet F1 (uncalibrated reference). $n = 1$ seed $\times$ 200 perturbations $\times$ 200 genes.

Magnitude calibration is necessary but insufficient. Calibration corrects ~40% of the substitutability gap (the marginal piece) and leaves ~60% (the joint piece) untouched. The 27–30% reversed-edge fraction in multi-hop analysis directly manifests joint-structure distortion: deep models connect the right gene pairs in the wrong direction, an error distributional calibration cannot fix because it is a property of conditional-independence structure. A direct test: add covariance-preservation (project predicted shifts onto the principal subspace of control covariance) and sparsity-injection (soft-threshold small effects to match the empirical near-zero fraction) modes, and measure gap closure.

STATE foundation model on K562. Routing the K562 200-gene subset through Arc Institute's SE-600M cell encoder and the ST-SE-Replogle zeroshot/k562 checkpoint (Arc Institute, 2025), STATE recovers $F_1 = 0.462$ (1061 edges; precision = 0.497, recall = 0.432) against real-data GIES ground truth at N=200 on the same overlay+GIES pipeline. This is the first model in the roster to materially exceed the ElasticNet baseline ($F_1 \approx 0.32$). The gap is consistent with the SE-600M backbone exposing structure unavailable to linear-on-controls regression, but absolute $F_1 < 0.5$: even a 2025 foundation model recovers fewer than half of real-data interventional edges. This refines, rather than overturns, the central claim — the substitutability gap holds for sub-foundation perturbation models (GEARS, CPA, Geneformer) but a sufficiently expressive backbone narrows it. We obtained this on a 320-cell subset (1 per perturbation + 150 controls) due to the ~28 s/cell SE-600M CPU cost; full-panel embedding was infeasible in the 6 h Modal budget (Appendix B.2).

Limitations. Six limitations are noted. Ground truth. Track A ground truth (real-data GIES) is internally consistent but does not establish biological correctness; a Track B evaluation against the literature-curated TRRUST v2 database (Han et al., 2018) is implemented in the released code but reported as a robustness check (Spearman concordance with Track A model rankings rather than absolute F1 against TRRUST). Pretraining contamination. Geneformer V2's Genecorpus pretraining includes Replogle 2022, so its Replogle F1 is partly memorization. We quantify this with a cell-line-holdout comparison (Geneformer F1 on K562 vs RPE1) rather than a temporally held-out post-pretraining dataset, which would require a 2024+ public Perturb-seq atlas on a Geneformer-corpus-excluded human cell type; no such dataset is currently available with $\geq$ 50 perturbation-eligible genes. Seed budget. Multi-seed CIs reported here use $n \leq 20$ seeds for K562 and $n = 1$ for RPE1 and Norman; the qualitative finding that pairwise differences among predictive methods are not significant is robust to seed count, but minimum detectable F1 differences at $n = 20$ are approximately 0.011. Cross-dataset deep-model scope. Deep-model training and evaluation on RPE1 and Norman is deferred to follow-up work; the cross-context and cross-paradigm claims in this paper rest on the ElasticNet baseline alone for those datasets (see Table 1). Model roster. GEARS, CPA, and Geneformer were chosen for architectural diversity at the time of pipeline construction; they predate STATE (Adduri et al., 2025; Arc Institute, 2025) and other 2025–2026 perturbation models. Recent benchmarks (Wong et al., 2025; Csendes et al., 2025; Wei et al., 2025; Viñas Torné et al., 2025) cover additional models on the per-gene prediction task but not on the causal-recovery task evaluated here. Gene-set scale. Primary results are at $N = 200$ for compute-tractability of GIES partitioning; scale-sweep numbers extend to $N = 200$ along seven points and inspre extends scaling to $\sim$1000 genes, but a head-to-head deep-model evaluation at N = 500–1000 is not yet completed.

6. Conclusion

PerturbCausal is the first benchmark testing whether virtual cell model predicted Perturb-seq data can substitute for real Perturb-seq in downstream causal gene regulatory network recovery. Across one primary dataset (Replogle K562) with up to 20 random seeds, five predictive methods (ElasticNet, CPA, GEARS, Geneformer, STATE), and four causal-discovery backends (GIES, inspre, PC, NOTEARS), no deep virtual cell model significantly exceeds a sparse linear regression baseline on causal-recovery F1; STATE narrows but does not close the gap. The substitutability failure decomposes into a magnitude marginal (mode collapse + effect-size inflation, the two equivalent views identified by Mejia et al. (2025) and our diagnostic, partially correctable post-hoc via Bolstad–Reinhard quantile matching) and a joint-distribution residual (covariance and edge-orientation structure that distributional calibration cannot reach); the benchmark, three-mode calibration diagnostic, multi-seed pipeline, and reproducibility infrastructure are released as the open-source PerturbCausal Python package.

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A. Additional results

A.1. Scale sweep precision across N=50–200

N (genes)	GT edges	ElasticNet	GEARS	CPA	Geneformer
50	169	0.345	0.280	0.302	0.292
75	309	0.309	0.351	0.344	0.316
100	509	0.344	0.339	0.335	0.348
125	646	0.356	0.360	0.326	0.349
150	821	0.368	0.310	0.347	0.394
175	1,002	0.369	0.401	0.378	0.340
200	1,217	0.378	0.392	0.385	0.373

Table 4. Scale-sweep precision against real-data GIES ground truth at seven gene-set sizes. Bold values indicate top performance per row.

A.2. Multi-hop credit at $k = 1, 2, 3$

Method	P@k=1	P@k=2	P@k=3	Boost (k=2)	Boost (k=3)
ElasticNet	0.425	0.540	0.543	+11.5	+11.9
GEARS	0.410	0.529	0.533	+11.8	+12.2
CPA	0.387	0.500	0.508	+11.3	+12.0
Geneformer	0.432	0.525	0.529	+9.2	+9.7
Overlay Real	0.423	0.533	0.539	+11.0	+11.6

Table 5. Multi-hop credit precision at hop-k over the N=200 K562 evaluation. Boost columns show percentage-point increase from k=1.

A.3. RPE1 cross-context structural comparison

Property	K562	RPE1
First singular vector variance	56.8%	34.5%
Top-5 singular variance	69.8%	68.5%
Frac near-zero effects	38.6%	15.4%
Frac large effects ($\lvert \log_2 \mathrm{FC} \rvert > 0.5$)	2.7%	17.3%
GIES edges (real data)	1,220	810

Table 6. Structural properties of real perturbation-response matrices for K562 and RPE1 at N=200.

A.4. Multi-seed bootstrap confidence

Across five seeds (42, 123, 456, 789, 1024) the four predictive methods sit within overlapping intervals on every metric (Table 7). The GRNBoost2 random-baseline floor confirms that all four predictive methods recover real signal substantially above chance, while none of the deep models significantly separates from ElasticNet at this seed budget.

A.5. Full PC and NOTEARS robustness tables

The main-text PC and NOTEARS Discussion paragraph cites summary statistics. Tables 8 and 9 report the full per-condition F1, precision, recall, and edge count against each backend's own real-data ground truth. Both backends produce sparser graphs than GIES at N=200 (PC: 682 edges, NOTEARS: 328 edges, vs. GIES: 1,220), and per-model F1s are uniformly small in absolute terms. Calibration zeros out NOTEARS recovery for all models because quantile matching pushes ACE entries below the w_{thresh}=0.1 weight threshold; under PC the calibrated and vanilla scores are within ±0.01 for all three models.

Method	F1	Precision	Recall	AUPRC
ElasticNet	0.426 ± 0.016	0.420 ± 0.014	0.431 ± 0.019	0.425 ± 0.016
CPA	0.412 ± 0.016	0.407 ± 0.013	0.417 ± 0.019	0.400 ± 0.022
GEARS	0.406 ± 0.015	0.404 ± 0.012	0.408 ± 0.018	0.419 ± 0.012
Geneformer	0.425 ± 0.011	0.422 ± 0.010	0.429 ± 0.013	0.422 ± 0.013
Overlay-real (oracle)	0.422 ± 0.009	0.420 ± 0.010	0.425 ± 0.009	0.420 ± 0.013
GRNBoost2 (random)	0.098 ± 0.004	0.074 ± 0.003	0.147 ± 0.005	0.077 ± 0.005

Table 7. Multi-seed mean and standard deviation across $n = 5$ seeds at N=200 on K562 against real-data GIES ground truth (1,220 edges per seed). Overlay-real is the same overlay-sampling pipeline applied to real perturbed cell predictions and serves as a noise-floor for the overlay step itself. GRNBoost2 is a structurally-distant baseline (gradient-boosted regression) included as a near-random reference.

Condition	Edges	F1	Precision	Recall
Real (PC ground truth)	682	—	—	—
GEARS-vanilla	556	0.0549	0.0612	0.0499
GEARS-calibrated	652	0.0480	0.0491	0.0469
CPA-vanilla	268	0.0800	0.1418	0.0557
CPA-calibrated	294	0.0779	0.1293	0.0557
Geneformer-vanilla	142	0.0097	0.0282	0.0059
Geneformer-calibrated	240	0.0043	0.0083	0.0029

Table 8. PC algorithm (Fisher-Z conditional independence at α=0.05) per-condition results on K562 ACE matrices at N=200, seed 42.

A.6. Six-variant CPA architectural ablation

To localize the substitutability gap to a specific architectural component, six CPA variants were trained, each modifying one architectural axis relative to the NB-loss baseline. Per-variant causal F1 is reported in Table 10. Spread across all six variants is 0.387–0.418 ($\Delta = 0.031$ F1), within the multi-seed CI of vanilla CPA. Removing the adversarial covariate disentangler is the single largest deletion (loses 0.025 F1, the largest drop), but it does not reproduce the gap to real-data GIES recovery. The collective implication is that the substitutability gap is not localizable to any one architectural component of CPA.

B. Supplementary methods

B.1. Virtual cell model hyperparameters

All four predictive models were trained on the same K562 200-gene subset using their authors' default hyperparameter values where possible; deviations are deliberate matches to the published reference configurations of each model.

B.2. STATE inference pipeline

The STATE result was obtained zero-shot from the released ST-SE-Replogle/zeroshot/k562 checkpoint via the state CLI v0.10.5. The pipeline has two stages: (1) SE-600M cell encoder (state emb transform) maps each input cell to a 2,058-dimensional X_{state} embedding; (2) the ST-SE-Replogle perturbation model (state tx infer) maps (X_{state}, perturbation) pairs to predicted gene-expression vectors over the same N=200 gene panel.

Reproducing the published checkpoint required diagnosing three distinct framework-versioning bugs:

1. Decoupled head dimension. The released checkpoint encodes a Llama backbone with hidden_size=328, num_attention_heads=12, and explicit head_dim=64, i.e. $328 \neq 12 \times 64 = 768$. Transformers $\geq$5 add a strict validate_architecture check that rejects hidden_size % num_attention_heads != 0 regardless of an explicit head_dim; transformers 4.45 accepts decoupled head dimensions but lacks the (2) compatibility we describe next; transformers 4.52.3 supports both. Fix: pin transformers==4.52.3 after pip install -e state.

2. is_causal kwarg pass-through. The state package's LlamaBidirectionalModel.forward sets is_causal=False

Condition	Edges	F1	Precision	Recall
Real (NOTEARS ground truth)	328	—	—	—
GEARS-vanilla	24	0.0511	0.3750	0.0274
GEARS-calibrated	0	0.0000	0.0000	0.0000
CPA-vanilla	18	0.0405	0.3889	0.0213
CPA-calibrated	0	0.0000	0.0000	0.0000
Geneformer-vanilla	7	0.0119	0.2857	0.0061
Geneformer-calibrated	0	0.0000	0.0000	0.0000

Table 9. NOTEARS (continuous-optimization with acyclicity constraint) per-condition results on K562 at N=200, seed 42, λ=0.05, weight threshold w_{thresh}=0.1.

Variant	Edges	F1	Precision	Recall
loss_gauss (Gaussian, not NB)	1,215	0.401	0.402	0.400
latent_16 (latent dim 16, base 64)	1,188	0.413	0.418	0.407
latent_256 (latent dim 256)	1,200	0.418	0.422	0.415
decoder_1layer (decoder depth 1)	1,207	0.418	0.420	0.416
layer_norm (LN replaces BN)	1,212	0.399	0.402	0.396
no_adversarial (reg=pen=0)	1,207	0.387	0.389	0.385

Table 10. Six-variant CPA architectural ablation on K562 at N=200, seed 42, partition size 20, against real-data GIES ground truth (1,220 edges). All other hyperparameters are held at the vanilla CPA baseline (n_latent=64, encoder layers = 3, decoder layers = 3, batch-norm, NB reconstruction loss, adversarial covariate disentanglement, max_epochs = 300).

and forwards it to the parent LlamaModel.forward via **flash_attn_kwargs. transformers 4.45's signature lacks the **flash_attn_kwargs catch-all, so the call raises TypeError: unexpected keyword argument 'is_causal'. transformers 4.52.3 introduces the catch-all and accepts the call.

3. ALL_PARALLEL_STYLES is None on Modal. In transformers 4.52.3 specifically, transformers.modeling_utils.ALL_PARALLEL_STYLES is initialized to None in some container configurations (verified locally and on Modal), making LlamaModel.__init__'s post_init hit if v not in ALL_PARALLEL_STYLES and raise TypeError: argument of type 'NoneType' is not iterable. Fix: during image build, replace the offending line in transformers/modeling_utils.py with if ALL_PARALLEL_STYLES is not None and v not in ALL_PARALLEL_STYLES:; this propagates to the state CLI subprocess, where a sitecustomize.py monkey-patch did not reliably fire.

The full set of attempted patches (14 attempts) and the final working Modal image specification is in scripts/modal_run_state_v3.py.

Subset rationale. STATE inference was run on a 320-cell subset of K562: 120 control cells (sampled from the ∼10,691 non-targeting cells) plus 200 perturbed cells (one per gene in the N=200 panel). The reduction is required because SE-600M wall-clock inference on Modal CPU instances measures ∼28 s/cell at batch_size=8; the full 38,979-cell K562 panel would take ∼300 h, which exceeds the 6-hour Modal function timeout. The 320-cell subset embeds in ∼13.5 minutes and produces 200 distinct STATE-predicted perturbation profiles—one per perturbation, the unit consumed by overlay sampling and downstream GIES.

B.3. Norman 2019 cross-paradigm pipeline

The Norman 2019 (Norman et al., 2019) dataset (GSE133344) was downloaded from CellxGene Census and contains 111,445 K562 cells under 237 unique perturbations (single and combinatorial). After dropping combinatorial ('A+B') perturbations and keeping only control (n=11,855 cells) plus single-gene perturbations with at least 10 cells, 102 eligible target genes remain, all measurable in the standard expression panel.

The substitutability evaluation re-uses the K562 protocol unchanged: 200 control cells + 100 cells per perturbation overlay-sampled from the ElasticNet predictions, vstacked into one expression matrix and fed to GIES with partition size 20 and seed 42.

Two practical engineering issues are noted. (i) multiprocessing.Pool(fork) for parallel partition execution dead-

Hyperparameter	Value
CPA (Lotfollahi et al., 2023)	
n_latent	64
n_layers_encoder	3
n_layers_decoder	3
batch_size	512
max_epochs	300
n_epochs_kl_warmup	20
n_epochs_adv_warmup	50
n_epochs_pretrain_ae	10
reconstruction loss	Negative Binomial
GEARS (Roohani et al., 2024)	
hidden_size	64
epochs	20
batch_size	32
Geneformer V2 (Theodoris et al., 2023)	
backbone	ctheodoris/Geneformer (HF)
fine-tuning task	in-silico perturbation
forward_batch_size	16
ElasticNet baseline (Zou & Hastie, 2005)	
alpha	0.1
l1_ratio	0.5
max_iter	1,000
STATE (Arc Institute, 2025)	
embedding model	SE-600M
prediction model	ST-SE-Replogle / zeroshot/k562
hidden_size	328
num_attention_heads	12
head_dim	64 (decoupled, $\neq H/h$)
num_hidden_layers	8
intermediate_size	3,072
max_position_embeddings	64
embedding dim (X_{state})	2,058

Table 11. Training hyperparameters per virtual cell model. Geneformer is fine-tuned from the public ctheodoris/Geneformer V2 checkpoint; STATE is run zero-shot from the released ST-SE-Replogle K562 checkpoint.

locks on the semaphore in Modal's container runtime; switching to sequential GIES execution removes the deadlock at the cost of single-thread CPU throughput (acceptable: total wall time ≈28 s on the M4). (ii) Some Norman partitions contain genes with zero variance under particular interventional regimes, producing 1/omega = inf in the gauss_int score and an effectively infinite hill-climbing loop in GIES. Adding $\mathcal{N}(0, 10^{-6})$ jitter to the partition expression matrix before scoring prevents the degeneracy without measurably altering recovered edge sets in the well-conditioned partitions.

B.4. Causal-discovery backend implementations

GIES (Hauser & Bühlmann, 2012): Greedy Interventional Equivalence Search via the gies Python package (v0.0.1), score function gauss_int_l0_pen, ∼20-gene partitions for tractability of the underlying CPDAG search. Output edges are the union of single-direction CPDAG arrows across partitions; bidirectional CPDAG arrows are kept once with the canonical ordering (lower index first).

inspre (Brown et al., 2025): Sparse-inverse regression on the perturbation-conditional ACE matrix via the inspre R package, full λ-path with leave-one-target-out cross-validation. The Modal R kernel (renv-isolated) is invoked via Rscript subprocess to keep the Python pipeline driver-only.

PC (Spirtes & Glymour, 2000): Constraint-based causal discovery via the causal-learn Python package, Fisher-Z conditional independence at $\alpha = 0.05$. Run on each model's per-perturbation ACE matrix.

NOTEARS (Zheng et al., 2018): Continuous-optimization with acyclicity constraint, $\lambda = 0.05$, weight threshold

$w_{\mathrm{thresh}} = 0.1$. Implementation: reference notears package on the same per-condition ACE inputs.

Overlay sampling (used by GIES, PC, NOTEARS): for each perturbed condition, 100 synthetic cells are produced by drawing 100 control cells with replacement, multiplying by a per-gene ratio of (predicted-mean $+10^{-2}$) over (control-mean $+10^{-2}$) clipped to $[0.01, 10]$, and Poisson-resampling to count-like expression. This preserves per-cell control covariance structure while imposing the predicted shift in marginal magnitude.

C. Reproducibility

C.1. Software environment

Component	Version
Python	3.9 (local) / 3.11 (Modal)
PyTorch	2.4.1 (CPU)
transformers	4.52.3 (STATE-pinned)
tokenizers	0.21.x
huggingface_hub	0.30.x
anndata	0.10.9
scanpy	1.10.3
numpy	1.26.4
scipy	1.13.1
scikit-learn	1.x
gies (Python)	0.0.1
causal-learn	0.1.x
notears	reference impl. from (Zheng et al., 2018)
inspre (R)	0.0.1
state (Arc Institute)	0.10.5

Table 12. Software versions used end-to-end. Modal containers pin all dependencies via the image build steps in scripts/modal_run_*.py.

C.2. Compute resources

Local pipeline (overlay sampling, GIES, PC, NOTEARS, calibration, multi-seed aggregation, plotting) runs on an Apple M4 laptop with 16 GB unified memory. Wall time: ≈45 minutes for the full N=200, single-seed K562 evaluation.

Modal cloud is used for the model-training and large-scale inference steps: GEARS, CPA, Geneformer, STATE, and the inspre N=500 scale point. Modal H100 instances are reserved for GEARS and Geneformer fine-tuning (≈1 h each at N=200); A10G for CPA training (≈45 minutes); 16-CPU 64 GB instances for inspre and the STATE CPU fallback. The largest single Modal job wall-clock is the SE-600M embedding step in the STATE pipeline at ≈3.5 hours for the full embedding, ≈13 minutes for the 320-cell subset.

C.3. Random seeds

The single-seed numbers throughout the main text use seed=42. Multi-seed CI tables in §A.4 use $\{42, 123, 456, 789, 1024\}$. Inside the GIES partitioning step, the gene-permutation seed is 42; partition contents are deterministic given the panel and seed.

C.4. End-to-end reproduction commands

All numbers in the main text and tables are reproducible from the released code with seed=42.

```
# Calibrate predictions
PYTHONPATH=. python3 scripts/calibrate_predictions.py \
    --diagnostics-out results/calibration_diagnostics.json

# Vanilla evaluation
```

```
PYTHONPATH=. OMP_NUM_THREADS=1 python3 \
  scripts/run_eval_local.py \
    --data-path data/k562_subset_n200_slim.h5ad \
    --pre-subset --n-genes 200 --partition-size 20 \
    --n-cells 100 --n-ctrl 200 --seed 42 \
    --model-outputs-dir data/model_outputs_real_n200 \
    --gies-cache-dir results/gies_cache --skip-random

# Calibrated evaluation
PYTHONPATH=. OMP_NUM_THREADS=1 python3 \
  scripts/run_eval_local.py \
    --data-path data/k562_subset_n200_slim.h5ad \
    --pre-subset --n-genes 200 --partition-size 20 \
    --n-cells 100 --n-ctrl 200 --seed 42 \
    --model-outputs-dir \
      data/model_outputs_real_n200_calibrated \
    --gies-cache-dir results/gies_cache_calibrated \
    --skip-random

# STATE evaluation (after Modal returns state_predictions.h5ad)
PYTHONPATH=. OMP_NUM_THREADS=1 python3 \
  scripts/state_eval_gies.py

# Norman 2019 cross-paradigm
.venv_gears/bin/python scripts/run_norman_local.py
```